

**JECRC UNIVERSITY, JAIPUR**  
**Session: Jan-June 2026**  
**Subject: Mathematics-II (DMA025A)**  
**B. Tech. II Semester I Year 2025-26**  
**Tutorial Sheet-1 (M.M: 64)**

**UNIT-1: Matrices**

**CO1:** The students will learn the essential tool of matrices and linear algebra in a comprehensive manner.

**PART – A (5\*2=10)**

**A1:** Define lower and upper triangular matrix.

**A2:** Find the value of x such that A is singular where  $A = \begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix}$  **Ans.:** 0, 3, 3

**A3:** Evaluate  $A^2 - 3A + 9I$  where  $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$  **Ans.:**  $\begin{bmatrix} -6 & 1 & 2 \\ 5 & 4 & 4 \\ 2 & 8 & -3 \end{bmatrix}$

**A4:** Define symmetric and skew-symmetric matrix.

**A5:** Define rank-nullity theorem.

**PART – B (3\*7=21)**

**B1:** Find the inverse of the following matrices by row transformation

$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$  **Ans.:**  $A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

**B2:** Solve the following homogeneous system of equations

$$x + 3y - 2z = 0, 2x - y + 4z = 0, x - 11y + 14z = 0$$

**Ans.:**  $-10k/7, 8k/7, k$  where k is any arbitrary constant.

**B3:** Find rank of the matrix  $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$  **Ans.:** 2

**PART – C (3\*11=33)**

**C1:** Test for consistency of the following system of equations and if possible, find the solution

$$x + y + z = 8, 2x + 3y + 2z = 19, 4x + 2y + 3z = 23$$

**Ans.:** 2, 3, 3

**C2:** Test for consistency of the following system of equations and if possible, find the solution

$$x + y + z = 6, x + 2y + 3z = 14, x + 4y + 7z = 30$$

**Ans.:**  $K-2, 8-2K, K$  where K is any arbitrary constant.

**C3:** Test for consistency of the following system of equations and if possible, find the solution

$$2x - 3y + 7z = 5, 3x + y - 3z = 13, 2x + 19y - 47z = 32$$

**Ans.:** System is inconsistent and there is no solution.

\*\*\*\*\*